

Series 1: Linear Models

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Applied Machine Learning and Predictive Modelling 1, FS26 (HSLU)

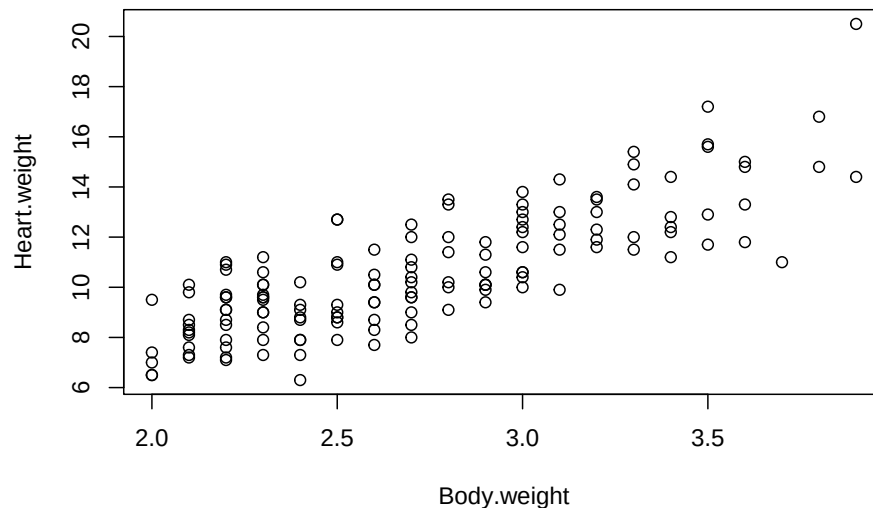
Exercise

In class we fitted a model to the “cats” dataset. You may remember that the interpretation of the intercept was somehow problematic. Let’s get the data, visualise it and refit the model again.

```
## (results are hidden)
d.cats <- read.csv("Cats.csv",
                  header = TRUE, stringsAsFactors = TRUE)
##
str(d.cats)
head(d.cats)
```

Let’s display the effect of *Body.weight*.

```
plot(Heart.weight ~ Body.weight, data = d.cats)
```



The first model we fitted was:

```
lm.cats <- lm(Heart.weight ~ Body.weight, data = d.cats)
```

The estimated coefficients of this model are:

```
coef(lm.cats)
```

```
(Intercept) Body.weight
-0.36      4.03
```

As mentioned in class, the correct interpretation of the intercept is “a cat with zero **body.weight**, is expected to have a heart weight of -0.36”. It is obviously nonsensical for two reasons: 1) there is no cat of zero body weight and 2) a negative prediction for the response variable **heart.weight** is impossible in reality.

Question

How would you proceed to simplify the interpretation of the intercept in this model? Hint: try to manipulate the predictor `body.weight` (e.g. by centering it).

Question

Let's turn our attention to the model that contains `sex` too, but no interaction. Reparametrise this model such that "M" is the reference. Hint: use the `relevel()` function.

Question

When the predictor `sex` was added to the model, the estimated coefficient for 'body.weight' slightly changed. Refit both models, show the estimated coefficients and write a sentence that correctly describes their "biological interpretation" of the `Body.weight` predictor in each model.

Question

This time we assume that `Body.weight` was not provided as a continuous variable, but rather as a categorical one. We do this by creating four classes with similar size. With this purpose in mind, we use the `quantile()` and `cut()` functions.

```
quantiles.Body.weight <- quantile(d.cats$Body.weight)
quantiles.Body.weight
```

```
0%  25%  50%  75% 100%
2.0  2.3  2.7  3.0  3.9
```

```
##
d.cats$Body.weight.Class <- cut(d.cats$Body.weight,
                                breaks = quantiles.Body.weight,
                                include.lowest = TRUE)
```

Let's check how many observations are present in each class.

```
table(d.cats$Body.weight.Class)
```

```
 [2,2.3]  (2.3,2.7] (2.7,3.02] (3.02,3.9]
      42         40         26         36
```

Fit a model with `Sex` and `Body.weight.Class` and compute a p-value for both predictors.

Question

Now run some contrasts to see whether all pair of levels of the `Body.weight.Class` predictor differ from each other. Comment on the results.

Question

Ask generative AI to provide the interpretation of

- the coefficients from a linear model
- the p-values from a linear model

and compare it with the definitions you find in the lecture materials. Do you think they are different in any way? Which one is easier for you to understand?